European Commission

Defence Industry and Space

Artificial Intelligence (AI) in Defence

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DEFENCE APPLICATIONS



Artificial Intelligence is rapidly evolving, with increasingly capable systems likely to be highly relevant for defence applications



AI could significantly enhance defence capability through improved threat detection and target identification, among others



Artificial Intelligence could be used for cyberattacks and intelligence breaches

Why does AI matter for defence?

Artificial Intelligence (AI) is a key driver in the current technological revolution in defence. AI global powers channel unprecedented resources into research and development because of the profound transformation it brings in modern warfare. AI's vast potential is reshaping military strategy, redefining operations, and revolutionising decision-making. The future of the **battlefield is as much about algorithms and compute power** as it is about arms.

Key differences between civil and defence AI

Differently from the civil context, defence systems must operate in adversarial high-risk scenarios, demanding extreme reliability, resilience and security. With human lives at stake, the costs of mistakes are high. This is why stringent testing and robust cybersecurity are at the core of European Defence Fund (EDF) projects. High quality, versatility, and a rigid testing regime yield a big spin-off potential outside the EDF, for example on drone technology and secure communications.

What does the EU aim to achieve?

Several AI-related projects funded by the **European Defence Fund (EDF)** aim to improve information superiority, operational effectiveness, interoperability, situational awareness, and AI-driven equipment design.

The EDF Indicative Multiannual Perspective 2025-2027 prioritises trustworthy AI for defence, focusing on enhancing AI performance in processing complex data (e.g. images, video, and text), improving human-machine interaction, advancing autonomous learning to enable AI systems to adapt with minimal human input, and developing energy-efficient AI hardware.

Main advantages of using AI in defence

- Real-time situational awareness
- Faster threat detection
- Better coordination of autonomous systems
- Optimised logistics and resources management
- Higher precision in target identification
- Reduced human deployment and exposure
- Improved battle damage assessment

Main vulnerabilities arising from the use of AI in defence

- Ethical and legal challenges
- Cyberattacks and adversarial AI
- Overreliance and reduced human oversight
- Unpredictability of AI in decision-making
- Data bias and reliability issues
- Inconsistent interoperability

When will EDF project results be translated into military capabilities?

Several projects are already delivering capabilities, with multiple currently ongoing projects expected to produce results in the coming years

The PANDORA project, which developed an **AI-enabled cyber defence platform** to be deployed within Member States' armed forces, is a successful example of an EU-funded defence research project. It focused on enhancing automated real-time incident response and information sharing and built on previous EU-funded defence research under EDIDP.

Other projects in the domain of AI for defence include **AInception** for network threat detection, **LODESTAR** for the integration of Augmented Reality and AI into future soldier systems, as well as **ARCHER, NEMO,** and **SALT4D** that deploy enhanced human language technologies in C4ISR systems. The EDF projects **FaRADAI, KOIOS,** and **ALADAN** are developing AI frameworks that learn faster and with less data.

Projects delivering capabilities currently in use

AIDEDex
AI-driven detection of explosive devices
EU financing: €5M

AI4DEF
AI cloud service platform
EU financing: €5.7M

PANDORA
AI-enabled Cyber Defence Platform
EU financing: €6.8M

PRIVILEGE
AI for military data encryption
EU financing: €1.4M

EDF projects expected to deliver results in 2026-2027

KOIOS
AI for military applications and simulation
EU financing: €10M

AInception
AI-based cyber intrusion detection tools
EU financing: €8.1M

EU-GUARDIAN
AI-based automation of cyber incident management
EU financing: €13.5M

FaRADAI
AI frameworks that learn faster with less data
EU financing: €18.5M



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